

ALDRIN R. VILLANUEVA

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| <https://github.com/ADrinnnu> | <https://aldrinportfolio.vercel.app>

TECHNICAL SKILLS

- **Frontend Development:** React.js (v18/19), Vite, Tailwind CSS, Zustand, TanStack Query, HTML5, CSS3, JavaScript.
- **Backend Development:** Node.js, Express.js, .NET (C#), Python, REST APIs.
- **Database & Cloud:** MySQL, PostgreSQL, AWS S3, MongoDB, Firestore.
- **Architecture & Security:** Clean Architecture, Role-Based Access Control (RBAC), JSON Web Tokens (JWT), bcrypt.
- **IT Operations & Support:** Hardware Troubleshooting, Network Configuration (Cisco), User Onboarding, and System Documentation.
- **Real-time & Tooling:** WebSockets (Socket.io, SignalR), Puppeteer (PDF Generation), Git/GitHub.

PROFESSIONAL EXPERIENCE & INCUBATION

Full Stack Developer Intern | Wireless Access for Health

February 2026 – May 2026

- Architected an ongoing full-stack Human Resources Information System (HRIS) and Payroll platform to automate the employee lifecycle for a multi-tenant user base.
- Engineered a secure document management pipeline using Node.js and AWS S3, enforcing strict Data Privacy through Role-Based Access Control (RBAC) and JWT session management.
- Developed a dynamic front end using React 19 and Zustand, implementing real-time live notifications for leave and offboarding approvals via Socket.io.
- Automated payroll reporting and attendance summaries by integrating Puppeteer for server-side PDF generation, significantly reducing HR administrative workload.

Full Stack Software Developer | PGT Transport Management System (University Incubation)

January 2026 – Present

- Developing an active, scalable logistics and Transport Management System (TMS) backend using .NET (C#) structured around Clean Architecture principles to ensure decoupled, enterprise-level code.
- Integrated React Google Maps API and Leaflet to construct interactive, geospatial interfaces for live route visualization and dispatch scheduling.
- Established real-time bi-directional data flow for live driver tracking and system status updates utilizing SignalR WebSockets.

NOTABLE PROJECTS

ChatFox: University AI Assistant | *React Native, Python, MongoDB, OpenAI API*

- Developed a cross-platform AI-powered chatbot for TSU to automate student inquiries and support
- Integrated a custom database with a Large Language Model (LLM) to ensure the AI provides accurate, institution-specific answers (RAG architecture).
- Reduced administrative workload by enabling students to instantly retrieve information on university guidelines and schedules.

Furniture E-Commerce Platform | React.js, Python, Firestore

- Developed a full-stack online shop with complete **CRUD functionality**, featuring a browsable **product catalog** where users can filter, view details, and purchase furniture.
- Implemented a personalized AI recommender system that suggests items based on user preferences.

Leadership & Activities

TSU Vulphox Esports

Committee Member

Tarlac City, Tarlac

Jan 2023 – Jan 2025

- Managed and organized esports tournaments during annual University Intramurals, coordinating schedules for multiple teams.
- Provided on-site technical support for gaming setups, including troubleshooting hardware connectivity and ensuring stable network performance during live matches.
- Collaborated with university staff to ensure smooth event logistics and quick resolution of technical glitches.

EDUCATION

Tarlac State University

Bachelor of Science in Information Technology

Specialize in Web and Mobile Applications

Relevant Coursework: Web Systems and Technologies; Operating Systems; Data Structures & Algorithms; Computer Networking; Database Management.

Tarlac City, Tarlac

Expected Graduation July 2026

CERTIFICATIONS

Cisco Certified Network Associate (CCNAv7) *Cisco Networking Academy*

- **Switching, Routing, and Wireless Essentials** (June 2024)
- **Introduction to Networks** (January 2024)

Career Essentials in System Administration *Microsoft & LinkedIn* (July 2024)

- Demonstrated proficiency in network security, user management, and server administration fundamentals.